

Sequential jump detection at high frequency via e-values and online false-discovery-rate control

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Abstract

This report documents the `efdr-jumps` project, whose goal is to detect price discontinuities (*jumps*) in high-frequency financial data in real time while simultaneously controlling the false discovery rate (*FDR*) in a sequential fashion. We construct *e-values* from the jump-robust volatility estimator MedRV (Andersen-Dobrev-Schaumburg, 2012) and feed them into five online FDR procedures from the Ramdas school (e-BH, e-LOND, e-LORD, e-SAFFRON, stopped e-BH). Monte-Carlo experiments (`grid_medium`, $M = 100$ replications, $n = 500$ steps, 3 frequencies, 3 jump regimes) show that the BH-based benchmarks (Yen, 2013; Bajgrowicz-Scaillet, 2016) violate the criterion $FDR \leq \alpha + 2 \text{ SE}$ at high frequency ($\Delta t = 5 \text{ s}$: observed $FDR = 0.096$ against the threshold 0.094 for $\alpha = 0.05$), whereas e-value-based procedures control the FDR at all frequencies, with a power loss of 10–21 percentage points.

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1 Introduction

1.1 Motivation: high-frequency price jumps

Modern financial markets generate transaction streams at millisecond frequencies. In this setting, the log-price $X_t = \log S_t$ is modelled as an Itô semimartingale with jumps:

$$dX_t = \mu_t dt + \sigma_t dW_t + \sum_k J_k dN_t^{(k)}, \quad (1)$$

where $\sigma_t > 0$ is the spot volatility, W_t a standard Brownian motion, and $J_k dN_t^{(k)}$ jump processes (compound Poisson or Hawkes). The jumps J_k represent sudden discontinuities induced by macroeconomic announcements, flash crashes, or liquidity events.

The presence of a jump at time t alters the local properties of the process drastically: the pure-diffusion assumption underlying most risk-management and pricing models is locally violated. Jump detection serves to clean data prior to calibration of stochastic-volatility models, to trigger alerts in market-surveillance systems, and to adjust positions in algorithmic execution strategies.

1.2 Problem statement: sequential detection with FDR guarantees

In discrete time, at each step $i = 1, \dots, n$, one observes the return $r_i = X_{i\Delta} - X_{(i-1)\Delta}$. We wish to test simultaneously n hypotheses:

$$H_{0,i} : r_i \sim \mathcal{N}(0, \Delta \sigma_i^2) \quad \text{vs.} \quad H_{1,i} : r_i = J_i + \varepsilon_i, \quad |J_i| \gg \sigma_i \sqrt{\Delta}. \quad (2)$$

with $n \sim 500$ for a trading day sampled at 5 s. The additional constraint is *online*: the decision on $H_{0,i}$ must be made at time i , without access to future observations.

1.3 State of the art

Barndorff-Nielsen & Shephard (2004, 2006). Bipower variation (BV) provides a jump-robust estimator of integrated variance. The BNS statistic tests globally for the presence of jumps on a window: it does not localize individual jumps.

Lee & Mykland (2008). The statistic $L_i = r_i / (\hat{\sigma}_i \sqrt{\Delta})$ is treated as $\mathcal{N}(0, 1)$ under $H_{0,i}$. BH [8] is then applied to two-sided p-values, enabling individual jump localization.

Yen (2013) and Bajgrowicz-Scaillet (2016). These works apply BH to Lee-Mykland p-values and to the BNS statistic respectively, providing the benchmarks reproduced in this project (`fdr/baselines.py`).

1.4 Core limitation: BH is invalid under HF dependence

BH guarantees $\text{FDR} \leq \alpha$ under the PRDS condition [9]. At high frequency ($\Delta t \leq 5$ s), the Lee-Mykland statistics normalized by BV exhibit a dependence structure that violates PRDS under dense jumps. Our central empirical finding confirms this limitation: at $\Delta t = 5$ s, BH-LM and BH-BNS produce $\text{FDR} \approx 0.096 > 0.094 = \alpha + 2 \text{SE}$ for $\alpha = 0.05$, $M = 100$ (source: `results/grid_medium.parquet`, aggregated across all regimes).

1.5 Project contributions

1. **E-value construction for HF jump detection:** integrated likelihood ratio under a Gaussian prior $\mathcal{N}(0, \tau^2)$ (`evalues/construct.py`), valid under arbitrary dependence.
2. **Implementation and validation of 7 FDR procedures:** e-BH, e-LOND, e-LORD, e-SAFFRON, stopped e-BH (e-values), and BH-LM, BH-BNS (baselines). 86 unit tests, all passing.
3. **Empirical evidence of BH's FDR violation at 5s** across 3 jump regimes (rare Merton, moderate Merton, dense Hawkes), 2 sizes (3σ , 5σ), 2 levels of α .
4. **Quantification of the power-safety trade-off:** 10–21 pp loss for guaranteed FDR control.
5. **Identification and correction of alpha-death** for e-LORD/e-SAFFRON: $\omega_1 = O(1/n)$ is required to avoid wealth exhaustion.

1.6 Outline

Section 2 lays out the mathematical framework (hypothesis testing, p/e-variables, multiple-testing metrics). Section 3 describes the simulated processes. Section 4 presents the volatility estimators. Section 5 details the e-value construction. Section 6 describes the seven FDR procedures with their proofs. Section 7 documents the software architecture. Section 8 reports Monte-Carlo results with figures. Section 9 discusses limitations. Section 10 concludes.

2 Theoretical framework

This section lays the foundations of multiple hypothesis testing used throughout the report. We start from first principles — hypotheses, tests, p- and e-variables — before turning to the multiple-testing setting and the FDR criterion. Procedure-specific definitions and proofs are deferred to Section 6.

2.1 Price model and definition of a jump

In discrete time with step Δt (in seconds), the log-price reads:

$$r_i = \underbrace{\mu \Delta}_{\text{drift}} + \underbrace{\sigma_i \sqrt{\Delta} Z_i}_{\text{diffusion}} + \underbrace{J_i \mathbb{1}_{\{\text{jump at } i\}}}_{\text{jump}}, \quad (3)$$

where $\Delta = \Delta t / \text{Seconds(per/year)}$ and $Z_i \sim \mathcal{N}(0, 1)$ i.i.d.

Definition 2.1 (Jump). $H_{0,i}$ (no jump at i): $r_i \mid \mathcal{F}_{i-1} \sim \mathcal{N}(0, \Delta \sigma_i^2)$. $H_{1,i}$ (jump at i): $r_i \mid \mathcal{F}_{i-1} \sim \mathcal{N}(J_i, \Delta \sigma_i^2)$ with $|J_i|$ large relative to $\sigma_i \sqrt{\Delta}$.

Convention : $\text{Seconds(per/year)} = 252 \times 6.5 \times 3600 = 5\,896\,800$, annualised volatilities, log-prices in natural log.

2.2 Hypotheses and tests

We adopt the formal framework of Ramdas & Wang (2025).

Definition 2.2 (Hypothesis). A *hypothesis* is a set of probability measures. It is *simple* if it is a singleton $\{P\}$, otherwise *composite*. We write $\mathcal{P}$ for the set of probability measures specifying the null hypothesis.

Definition 2.3 (Test). A *test* ϕ is a $[0, 1]$ -valued random variable measurable with respect to the data σ -algebra. It is called *pure* (or *binary*) if $\phi \in \{0, 1\}$ almost surely.

For a simple null P , the *type-I error rate* is $\mathbb{E}^P[\phi]$; for a composite null $\mathcal{P}$, it is $\sup_{P \in \mathcal{P}} \mathbb{E}^P[\phi]$. A test has *level* $\alpha \in [0, 1]$ if its type-I error rate is at most α .

Let $\mathcal{M}$ be the set of all probability measures on $(\Omega, \mathcal{F})$ and let $H_1 = \mathcal{M} \setminus \mathcal{P}$ be the alternative.

Definition 2.4 (Power and unbiasedness). The *power* of a test against the alternative is the map $Q \mapsto \mathbb{E}^Q[\phi]$ for $Q \in H_1$. The test is *unbiased at level α* against H_1 if for every $P \in \mathcal{P}$ and every $Q \in H_1$:

$$\mathbb{E}^P[\phi] \leq \alpha, \quad \mathbb{E}^Q[\phi] \geq \alpha.$$

For binary tests $\phi = \mathbb{1}_{\{\text{reject}\}}$, the *type-II error rate* is $1 - \mathbb{E}^Q[\phi]$.

2.3 P-variables, e-variables, and calibrators

Definition 2.5 (P-variable and e-variable). A *p-variable* for $\mathcal{P}$ is a $[0, 1]$ -valued random variable P satisfying $\mathbb{P}(P \leq \alpha) \leq \alpha$ for every $\alpha \in [0, 1]$ and every $P \in \mathcal{P}$.

An *e-variable* for $\mathcal{P}$ is a non-negative random variable E satisfying $\mathbb{E}^P[E] \leq 1$ for every $P \in \mathcal{P}$. It is *exact* if equality holds.

We use “variable” for the random object and “value” for its realisation. Two key results connect these two objects.

Theorem 2.6 (E-to-p calibration). *Let E be an e-variable for $\mathcal{P}$. Then for every $\alpha \in [0, 1]$ and every $P \in \mathcal{P}$:*

$$\mathbb{P}(E \geq 1/\alpha) \leq \alpha.$$

Hence $P := \min(1/E, 1)$ is a p-variable, and the map $e \mapsto \min(1/e, 1)$ is the (essentially unique) admissible e-to-p calibrator.

Proof. Markov’s inequality applied to the non-negative random variable E . □

Thus rejecting H_0 when $E \geq 1/\alpha$ defines a level- α binary test. The converse is *false*: a p-variable is not in general an e-variable (Vovk & Wang, 2021).

Definition 2.7 (P-to-e calibrator). A *p-to-e calibrator* is a decreasing function $f : [0, 1] \rightarrow [0, +\infty]$ such that $f(P)$ is an e-variable for every p-variable P . A calibrator f *dominates* g if $f \geq g$; it is *admissible* if not dominated by another calibrator.

Remark 2.8 (Vovk-Wang calibrator). For $\kappa \in (0, 1)$, the map $p \mapsto \kappa p^{\kappa-1}$ is an admissible p-to-e calibrator: for any p-variable P , the random variable $E_\kappa = \kappa P^{\kappa-1}$ is an exact e-variable. In this project, $\kappa = 0.5$ serves as the baseline in validity tests (`test_fdr_algorithms.py`); the GROW construction is required to dominate this calibrator in e-power.

The motivation for using e-variables, beyond their richer calibration theory, is their robustness to *dependence*: many e-value-based procedures require no independence or PRDS assumption, whereas most p-value procedures do. We close with the notion of admissibility.

Definition 2.9 (Admissible e-variable). An e-variable E for $\mathcal{P}$ is *admissible* if any e-variable $E' \geq E$ (P -almost surely, for every $P \in \mathcal{P}$) satisfies $E' = E$ P -a.s.

2.4 Multiple testing: metrics and the price of safety

Consider a stream of hypotheses $H_1, \dots, H_K$ (K possibly infinite). In the modern multiple-testing framework, we view the data as generated by a single joint law P , and each test provides evidence about a feature of P . Formally, $H_{0,k} : P \in \mathcal{P}_k$ where $\mathcal{P}_k$ is a subset of measures. We write

$$H_0(t) = \{k \in \{1, \dots, t\} : P \in \mathcal{P}_k\}$$

for the set of true null indices up to time t , and $\mathcal{K} = \{1, \dots, K\}$.

Definition 2.10 (Multiple testing procedure). Let X denote the available data with joint law P . A *multiple testing procedure* D is a Borel function of X returning a subset of $\mathcal{K}$, interpreted as the indices of rejected hypotheses. A procedure taking e-values as input is an *e-testing procedure*.

Write $R_D = \text{Card}(D)$ for the number of rejections and

$$F_D = \sum_{k \in H_0} \mathbb{1}_{\{k \in D\}}$$

for the number of *false discoveries* (rejected true nulls).

Definition 2.11 (Error metrics).

$$\begin{aligned} \text{FDP} &= \frac{F_D}{R_D \vee 1} && \text{(false discovery proportion),} \\ \text{FDR} &= \mathbb{E}^P[\text{FDP}] && \text{(false discovery rate),} \\ \text{FWER} &= \mathbb{P}(F_D \geq 1) && \text{(family-wise error rate),} \\ \text{mFDR}(\eta) &= \frac{\mathbb{E}^P[F_D]}{\mathbb{E}^P[R_D] + \eta} && \text{(marginal FDR),} \end{aligned}$$

with the convention $0/0 = 0$ encoded by $R_D \vee 1$ in the denominator.

A procedure has *FDR control at level α* if $\text{FDR} \leq \alpha$ for every $P \in \mathcal{P}$, and similarly for FWER.

Why FDR, not FWER. A naive procedure testing each hypothesis at level α produces, in expectation, $K\alpha$ false discoveries: if K is large, false discoveries are inevitable. The classical Bonferroni correction sets $\alpha_k = \alpha/K$ and rejects H_k when $E_k \geq K/\alpha$. By a union bound:

$$\mathbb{P}(F_D \geq 1) \leq \sum_{k \in H_0} \mathbb{P}(\text{reject } H_k) \leq K_0 \cdot \frac{\alpha}{K} \leq \alpha,$$

giving FWER control. (Note: the quantity bounded above by $K\alpha$ is the *per-family error rate* $\mathbb{E}^P[F_D]$, distinct from FWER.) FWER, however, becomes vacuous for large K : requiring *no*

false discoveries is too strong when one expects many true alternatives. The FDR criterion, introduced by Benjamini-Hochberg [8], controls the *proportion* of false discoveries, allowing some at the price of meaningful power.

The cost of FDR control. Tighter control reduces power. The trivial procedure that rejects nothing has FDR 0 and power 0; the goal is therefore to *maximize expected true discoveries subject to* $\text{FDR} \leq \alpha$. Comparing procedures controlling different metrics (e.g. Bonferroni for FWER vs. e-BH for FDR) is not meaningful, since they solve different problems.

2.5 Offline vs. online: the α -investing framework

A procedure is *offline* when all hypotheses are available simultaneously: one may sort, threshold, and revise decisions globally (Bonferroni and BH are offline). It is *online* when hypotheses arrive in a stream, decisions are taken on the fly, and past decisions cannot be revised; the number of hypotheses may be infinite.

In the online setting, sorting is impossible and one needs a *wealth process* to allocate the type-I budget over time — the α -*investing* metaphor. Start with initial wealth ω_0 ; at each step t allocate a fraction α_t of the current wealth as the rejection threshold of the t -th test. The allocation α_t is spent regardless of the outcome; if the test rejects, a reward β_t is collected:

$$\omega_{t+1} = \omega_t - \alpha_t + \beta_t \mathbb{1}_{\{t \in D\}}. \quad (4)$$

The full sequence (α_t) is chosen so that the (running) FDR remains bounded by α for all t , often via a *predictable* construction based on $\mathcal{F}_{t-1}$. This framework is the basis of e-LOND, e-LORD, and e-SAFFRON (Section 6).

2.6 E-power and GROW optimality

For sequential procedures, type-II error has a refined counterpart: how fast does E grow under the alternative?

Definition 2.12 (E-power, Grünwald-de Heide-Koolen 2024).

$$\text{e-power}(E) = \mathbb{E}_{H_1}[\log E].$$

The e-power governs the rate of wealth accumulation in α -investing procedures: an e-power > 0 under H_1 is the efficiency condition for sequential tests. Among all e-variables satisfying the validity constraint $\mathbb{E}_{H_0}[E] \leq 1$, the one maximising e-power under a fixed prior π on the alternative is the *growth-rate optimal in the worst case* (GROW) e-variable.

Definition 2.13 (GROW e-value). Given a prior π on the alternative parameter (here the jump size J), the GROW e-value is

$$E^{\text{GROW}} = \int \frac{dP_J}{dP_0}(r_i) d\pi(J),$$

the marginal likelihood ratio under the mixture. It maximises e-power subject to $\mathbb{E}_{H_0}[E] \leq 1$.

2.7 Roadmap of procedures

The seven procedures implemented and benchmarked in this project are summarised in Table 1. Definitions and proofs are given in Section 6.

Table 1: Procedures implemented in `efdr-jumps`.

Procedure	File	Online	Dependence	Reference
e-BH	<code>fdr/ebh.py</code>	No	Arbitrary	Wang-Ramdas 2022
BH-LM	<code>fdr/baselines.py</code>	No	PRDS	Lee-Mykland 2008
BH-BNS	<code>fdr/baselines.py</code>	No	PRDS	BS 2016
e-LOND	<code>fdr/elond.py</code>	Yes	Arbitrary	Xu-Ramdas 2024
e-LORD	<code>fdr/elord_esaffron.py</code>	Yes	Arbitrary	Zhang et al. 2025
e-SAFFRON	<code>fdr/elord_esaffron.py</code>	Yes	Arbitrary	Zhang et al. 2025
Stopped e-BH	<code>fdr/stopped_ebh.py</code>	Yes	Markov / causal	Wang et al. 2025

3 Data generation

3.1 Heston model (stochastic volatility)

The instantaneous variance V_t under the Heston model follows:

$$dV_t = \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dW_t^V, \quad (5)$$

$$d \log S_t = (\mu - V_t/2) dt + \sqrt{V_t} dW_t^S, \quad (6)$$

with $d\langle W^S, W^V \rangle_t = \rho dt$.

Parameters (`simulate/heston.py`, lines 89–106):

$$\kappa = 2.0, \quad \theta = 0.04, \quad \xi = 0.5, \quad \rho = -0.7, \quad V_0 = 0.04, \quad \mu = 0. \quad (7)$$

Hence $\sigma_\infty = \sqrt{\theta} = 0.20$ (20% annualised). The leverage effect $\rho = -0.7$ reproduces the observed price-volatility correlation.

Andersen’s QE scheme (2007). The naive Euler-Maruyama scheme produces negative variances for Heston. The Quadratic Exponential (QE) scheme [4] models $V_{t+\Delta} \mid V_t$ as a mixture switching on the conditional dispersion $\psi = \text{Var}[V_{t+\Delta} \mid V_t] / (\mathbb{E}[V_{t+\Delta} \mid V_t])^2$:

- $\psi \leq 1.5$: lognormal approximation.
- $\psi > 1.5$: Dirac-exponential mixture.

This bifurcation guarantees $V_t \geq 0$ without ad-hoc correction. Implementation: `simulate/heston.py`, `_qe_kernel`.

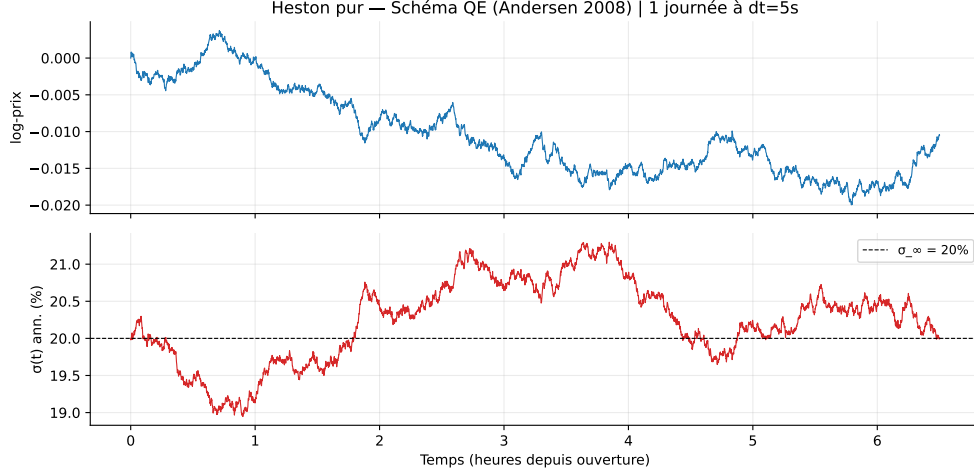


Figure 1: Simulated Heston log-price path with stochastic volatility ($\kappa = 2$, $\theta = 0.04$, $\xi = 0.5$, $\rho = -0.7$). The QE scheme guarantees $V_t \geq 0$.

3.2 Merton model (compound Poisson jumps)

The log-price follows the dynamics with compound Poisson jump process:

$$d \log S_t = (\mu - V_t/2 - \lambda \bar{k}) dt + \sqrt{V_t} dW_t^S + J dN_t, \quad (8)$$

where $N_t \sim \text{Poisson}(\lambda)$, $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$, and $\bar{k} = e^{\mu_J + \sigma_J^2/2} - 1$ corrects the drift so that S_t is a martingale (Andersen-Benzoni-Lund 2002 parameterisation, `simulate/merton.py`).

Grid parameterisation. Jump size is specified as a multiple of the local volatility (`pipeline/detector.py`):

$$\mu_J = \text{jump_size_sigma} \times \sqrt{\theta} \times \sqrt{\Delta}, \quad \sigma_J = 0.1 \times |\mu_J|. \quad (9)$$

For `jump_size_sigma = 5` and $\Delta t = 5$ s: $\mu_J \approx 9.21 \times 10^{-4}$ (in log-price), a jump of size $5\sigma\sqrt{\Delta}$. Jump intensities per regime (`detector.py`):

$$\lambda_{\text{rare}} = 1\,500/\text{year} \approx 0.64 \text{ jumps/path}, \quad \lambda_{\text{moderate}} = 5\,000/\text{year} \approx 2.12 \text{ jumps/path} \quad (10)$$

for $n = 500$, $\Delta t = 5$ s.

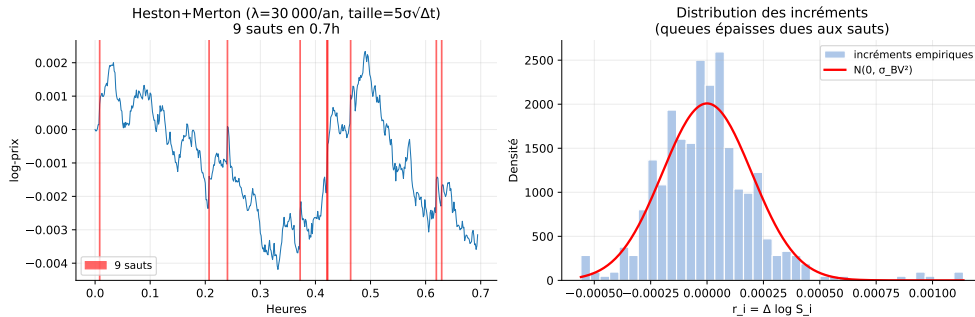


Figure 2: Distribution of Merton jump sizes under the three regimes (rare, moderate, hawkes_dense). Jump sizes are specified as multiples of the local volatility $\hat{\sigma}_i\sqrt{\Delta}$.

3.3 Self-exciting Hawkes process

To model the temporal dependence of jumps, a discrete Hawkes process is used (`simulate/hawkes.py`, 104 l.):

$$\lambda_{n+1} = \mu + (\lambda_n - \mu) e^{-\beta \Delta t} + \alpha N_n, \quad (11)$$

where $N_n \in \{0, 1\}$ and $p_n = 1 - e^{-\lambda_n \Delta}$ (`hawkes.py`, line 85).

Parameters (`simulate/hawkes.py`, lines 43–57):

$$\mu = 5\,000/\text{year}, \quad \alpha = 88\,000/\text{year}, \quad \beta = 0.03/\text{s}. \quad (12)$$

Branching ratio: $m = \alpha/(\beta \times \text{Seconds}(\text{per/year})) = 88\,000/(0.03 \times 5\,896\,800) \approx 0.497 < 1$. An error is raised if $m \geq 1$ (`hawkes.py`, lines 59–65).

Validation. Over 30 seeds ($n = 200$, $\Delta t = 5$ s): 3.8 jumps/path vs. 1.3 for `moderate` (ratio ≈ 3).

Note on discretisation. At $\Delta t = 300$ s, the approximation of the continuous Hawkes process by this discrete scheme degrades. The process remains well-defined ($p_n \leq 1$) but the low-frequency results for the `hawkes_dense` regime should be interpreted with care.

3.4 Discretisation convention

$\text{Seconds}(\text{per/year}) = 252 \times 6.5 \times 3600 = 5\,896\,800$. Returns r_i are in natural log. Δ is in years. The `jump_indices` provided by the simulator are indices into the log-price array; the corresponding return index is $j - 1$ (`detector.py` line 154).

4 Spot volatility estimation

4.1 Bipower variation (BNS 2004)

Definition 4.1 (BV).

$$\text{BV}_n = \frac{\pi}{2} \cdot \frac{1}{n-1} \sum_{i=2}^n |r_{i-1}| |r_i|. \quad (13)$$

Under pure diffusion, $\text{BV}_n \rightarrow_P \int_0^T \sigma_t^2 dt$. Under jumps, BV is downward-biased: the products $|r_{i-1}| |r_i|$ are close to zero when one of the returns carries an isolated jump.

4.2 MedRV and MinRV (ADS 2012)

Definition 4.2 (MedRV, ADS 2012 eq. 12).

$$\text{MedRV}_n = c_{\text{med}} \cdot \frac{n}{n-2} \sum_{i=2}^{n-1} \text{med}(|r_{i-1}|, |r_i|, |r_{i+1}|)^2, \quad c_{\text{med}} = \frac{\pi}{6 - 4\sqrt{3} + \pi} \approx 1.4191. \quad (14)$$

Definition 4.3 (MinRV, ADS 2012 eq. 13).

$$\text{MinRV}_n = c_{\min} \cdot \frac{n}{n-1} \sum_{i=2}^n \min(|r_{i-1}|, |r_i|)^2, \quad c_{\min} = \frac{\pi}{\pi-2} \approx 2.7141. \quad (15)$$

Implemented in `estimators/threshold.py` (43 l.).

4.3 Local spot volatility and causality condition

`estimators/spot.py` (76 l.) computes $\hat{\sigma}^2(i)$ over a window of K returns *strictly preceding* i :

$$\hat{\sigma}^2(i) = \hat{V}(\log S[\max(0, i-K) : i+1], \Delta t). \quad (16)$$

The return r_i is **never** included in the estimation window — this is the *causality condition* required for $\hat{\sigma}_i$ to be $\mathcal{F}_{i-1}$ -measurable, ensuring $\mathbb{E}[E_i | \mathcal{F}_{i-1}] \leq 1$.

The window is $K = 50$ in the pipeline (`EstimatorConfig.window = 50`, `detector.py` line 71). The first K hypotheses are excluded from the test (incomplete window).

Classical test statistics. $L_i = r_i / (\hat{\sigma}_i \sqrt{\Delta}) \approx \mathcal{N}(0, 1)$ under $H_{0,i}$ (Lee-Mykland), two-sided p-value $P_i = 2\mathbb{P}(|Z| > |L_i|)$. Implemented in `estimators/bipower.py` via `lee_mykland_stat`.



Figure 3: Comparison of spot volatility estimators (BV, MedRV, MinRV) on a simulated Heston path with Merton jumps. MedRV is the most robust to jump contamination.

5 E-value construction

5.1 Local test formulation

For each $H_{0,i}$:

$$H_{0,i} : r_i \mid \mathcal{F}_{i-1} \sim \mathcal{N}(0, v), \quad v = \Delta \hat{\sigma}_i^2, \quad (17)$$

$$H_{1,i}(J) : r_i \mid \mathcal{F}_{i-1} \sim \mathcal{N}(J, v) \quad (18)$$

for an unknown jump size J .

5.2 Integrated likelihood ratio under a Gaussian prior

The GROW construction chooses $\pi(J) = \mathcal{N}(0, \tau^2)$ and defines:

$$E_i = \int_{-\infty}^{+\infty} \exp\left(\frac{Jr_i}{v} - \frac{J^2}{2v}\right) \cdot \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{J^2}{2\tau^2}\right) dJ. \quad (19)$$

The derivation (Appendix A) gives the closed form:

$$E_i = \sqrt{\frac{v}{v + \tau^2}} \cdot \exp\left(\frac{r_i^2 \tau^2}{2v(v + \tau^2)}\right), \quad v = \Delta \hat{\sigma}_i^2, \quad \tau = 5 \hat{\sigma}_i \sqrt{\Delta}. \quad (20)$$

Parameter tau_factor = 5. $\tau = 5 \hat{\sigma}_i \sqrt{\Delta}$ places mass on jumps of size $\pm 5 \hat{\sigma}_i \sqrt{\Delta}$, consistent with the grid regimes (3σ and 5σ).

5.3 Validity: $\mathbb{E}[E_i \mid \mathcal{F}_{i-1}] = 1$ under $H_{0,i}$

Proposition 5.1. *If $\hat{\sigma}_i$ is $\mathcal{F}_{i-1}$ -measurable and $r_i \mid \mathcal{F}_{i-1} \sim \mathcal{N}(0, v)$ under $H_{0,i}$, then $\mathbb{E}[E_i \mid \mathcal{F}_{i-1}] = 1$.*

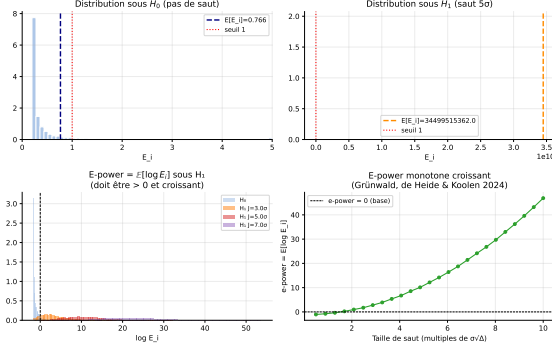
Proof. $E_i = \int (dP_J/dP_0)(r_i) d\pi(J)$ is the marginal density under the mixture divided by P_0 . By Fubini: $\mathbb{E}_{P_0}[E_i] = \int \mathbb{E}_{P_0}[dP_J/dP_0] d\pi(J) = \int 1 d\pi(J) = 1$. $\square$

The measurability condition is ensured by excluding the tested point from the window (Section 4.3).

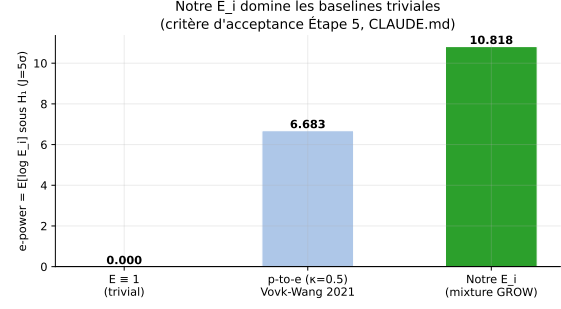
5.4 Empirical verification

Four unit tests (`test_evalues_validity.py`, 223 l.) verify:

1. **H0/H1 separation:** distribution of E_i concentrated near 1 under pure diffusion, shifted toward large values under a jump.
2. **Monotone e-power:** $\mathbb{E}[\log E_i]$ strictly increasing in jump size (from 3σ to 7σ).
3. **Dominance over p-to-e:** e-power of the GROW construction exceeds that of the $\kappa = 0.5$ calibrator.
4. **Validity:** $\mathbb{E}[E_i] \leq 1 + 3 \text{SE}$ under H_0 , $M = 10\,000$.



(a) Distribution of E_i under H_0 and H_1 .



(b) E-power: GROW vs. Vovk-Wang calibrator ($\kappa = 0.5$).

Figure 4: Empirical validation of the GROW e-value construction. Left: under H_0 the distribution is concentrated near 1; under H_1 it is shifted to large values. Right: the GROW construction dominates the baseline calibrator in e-power across jump sizes.

6 Online FDR control procedures

This section presents the seven procedures benchmarked in Section 8. The exposition follows the offline to online progression set up in Section 2.4: e-BH is offline, e-LOND, e-LORD, e-SAFFRON, and stopped e-BH are online. Proofs of FDR control rely on the e-value framework of Section 2.3.

6.1 e-BH (Wang & Ramdas 2022)

The e-BH procedure is the e-value analogue of Benjamini-Hochberg. Before stating it, we need the notion of *compound e-variables*, which extends e-variables to sets of indices.

Definition 6.1 (Compound e-variables). Let $H_1, \dots, H_K$ be null hypotheses and $E_1, \dots, E_K$ non-negative random variables. They are *compound e-variables* for $(H_1, \dots, H_K)$ if

$$\mathbb{E} \left[\sum_{k \in N} E_k \right] \leq |N|$$

for every configuration of true nulls $N \subseteq [K]$. They are *tight* if equality holds for some N .

A collection of K e-variables is automatically a collection of compound e-variables, but the converse is false in general.

Definition 6.2 (e-BH, Wang & Ramdas 2022). Let $e_1, \dots, e_K$ be realised compound e-values and let $e_{[k]}$ denote the k -th order statistic from the largest to the smallest. The e-BH procedure at level $\alpha \in (0, 1)$ rejects the hypotheses corresponding to the largest k^* e-values, where

$$k^* := \max \left\{ k \in \{1, \dots, K\} : \frac{k e_{[k]}}{K} \geq \frac{1}{\alpha} \right\} \quad (k^* := 0 \text{ if the set is empty}).$$

If $k^* = 0$, no hypothesis is rejected. The complexity is $O(K \log K)$ for sorting, implemented in `fdr/ebh.py` (44 l.).

Theorem 6.3 (Wang & Ramdas 2022). *For arbitrary dependence among the e -values, the e -BH procedure at level α satisfies*

$$\text{FDR} \leq \alpha \cdot \frac{|N|}{K} \leq \alpha.$$

The proof factors through a generic property called *self-consistency*, which characterises a broad class of FDR-controlling procedures.

Definition 6.4 (Self-consistency). An e -testing procedure D is *self-consistent at level α* if for every $k \in D$:

$$e_k \geq \frac{K}{\alpha R_D}.$$

Proposition 6.5. *The e -BH procedure is self-consistent. Moreover, any self-consistent e -testing procedure at level α has $\text{FDR} \leq \alpha K_0/K$ for arbitrary configurations of e -values, where $K_0 = |N|$.*

Proof. Let D be a self-consistent e -testing procedure, and let $E = (E_1, \dots, E_K)$ be the input e -variables. By self-consistency, for every $k \in D$, $E_k R_D \geq K/\alpha$, hence on the event $\{R_D \geq 1\}$:

$$\frac{\mathbb{1}_{\{k \in D\}}}{R_D} \leq \frac{\alpha E_k}{K}.$$

Therefore the FDP of D satisfies

$$\frac{FD}{R_D \vee 1} = \frac{|D(E) \cap N|}{R_D \vee 1} \leq \sum_{k \in N} \frac{\mathbb{1}_{\{k \in D(E)\}} \alpha E_k}{K} \leq \sum_{k \in N} \frac{\alpha E_k}{K}.$$

The first inequality applies self-consistency conditionally on $k \in D$. Taking expectations:

$$\mathbb{E} \left[\frac{FD}{R_D \vee 1} \right] \leq \sum_{k \in N} \frac{\alpha \mathbb{E}[E_k]}{K} \leq \frac{\alpha K_0}{K},$$

since $\mathbb{E}[E_k] \leq 1$ for $k \in N$.

The e -BH procedure is self-consistent by construction: for $k \in D$, $e_k \geq e_{[R_D]} \geq K/(\alpha R_D)$ by definition of $k^* = R_D$. $\square$

Theorem 6.3 follows by Proposition 6.5 applied to e -BH.

6.2 e -LOND (Xu & Ramdas 2024)

The online setting requires a different machinery: e -values arrive in a stream $(E_t)_{t \geq 1}$ and decisions must be made on the fly. We work with the natural filtration $\mathcal{F}_t = \sigma(E_1, \dots, E_t)$, $\mathcal{F}_0$ trivial.

Definition 6.6 (e -process). An e -process is a non-negative adapted sequence $(E_t)_{t \geq 0}$ such that for every stopping time τ w.r.t. $(\mathcal{F}_t)$, $\mathbb{E}[E_\tau] \leq 1$.

Since FDR is by definition the expectation of a proportion whose numerator and denominator may both be unbounded as $t \rightarrow \infty$, we work with the running quantity $\text{FDR}(t)$.

Definition 6.7 (e-LOND, Xu & Ramdas 2024). At step $t \geq 1$, set

$$\alpha_t = \alpha \cdot \gamma_t \cdot (R_{t-1} + 1), \quad \gamma_t = C_{\text{JM}} \cdot \frac{\log \max(t, 2)}{t \cdot \exp \sqrt{\log \max(t, 2)}}, \quad (21)$$

where R_{t-1} is the number of rejections before time t and $(\gamma_t)_{t \geq 1}$ is a user-chosen non-increasing non-negative sequence satisfying $\sum_{t \geq 1} \gamma_t \leq 1$ (the choice above is the Javanmard-Montanari sequence). Reject H_t iff $E_t \geq 1/\alpha_t$.

Constant $C_{\text{JM}} = 0.15708906$. Verified by truncated summation up to $t = 10^6$: $\sum_{t=1}^{10^6} \gamma_t \approx 1.000000$ (fdr/elond.py).

Interpretation. The budget $\alpha \sum_t \gamma_t \leq \alpha$ is distributed over time. The factor $(R_{t-1} + 1)$ raises the rejection threshold after each rejection (wealth accumulation). The decay $\gamma_t \sim \log(t)/(t e^{\sqrt{\log t}})$ ensures validity for infinite streams.

Theorem 6.8 (Xu & Ramdas 2024). *e-LOND controls $\text{FDR} \leq \alpha$ under arbitrary dependence.*

6.3 e-LORD (e-GAI framework, Zhang et al. 2025)

e-LORD takes a more flexible route via an *oracle FDP estimator*. The decision rule is again $\delta_t = \mathbb{1}_{\{E_t \geq 1/\alpha_t\}}$, but α_t is now built from an estimator that adapts to the rejection history.

Define the (unobservable) *oracle FDP*:

$$\text{FDP}^*(t) = \sum_{j \in H_0(t)} \frac{\alpha_j}{R_{j-1} + 1}.$$

It is called *oracle* because the summation runs over true nulls $H_0(t)$, which are unknown. The corresponding (observable) e-LORD estimator sums over all hypotheses:

$$\widehat{\text{FDP}}_{\text{e-LORD}}(t) = \sum_{j=1}^t \frac{\alpha_j}{R_{j-1} + 1}.$$

Proposition 6.9. *If $\mathbb{E}[\text{FDP}^*(t)] \leq \alpha$ for all $t \geq 1$, then $\text{FDR}(t) \leq \alpha$ for all $t \geq 1$.*

Proof. Bounding the indicator $\delta_j = \mathbb{1}_{\{E_j \geq 1/\alpha_j\}}$ by $\alpha_j E_j$ (Markov), and using $R_t \vee 1 \geq R_{j-1} + 1$ for $j \leq t$ contributing to R_t :

$$\begin{aligned} \text{FDR}(t) &= \mathbb{E} \left[\frac{\sum_{j \in H_0(t)} \delta_j}{R_t \vee 1} \right] \leq \mathbb{E} \left[\sum_{j \in H_0(t)} \frac{\delta_j}{R_{j-1} + 1} \right] \\ &= \mathbb{E} \left[\sum_{j \in H_0(t)} \frac{\mathbb{1}_{\{E_j \geq 1/\alpha_j\}}}{R_{j-1} + 1} \right] \leq \mathbb{E} \left[\sum_{j \in H_0(t)} \frac{E_j \alpha_j}{R_{j-1} + 1} \right] \\ &= \mathbb{E} \left[\sum_{j \in H_0(t)} \frac{\mathbb{E}[E_j \mid \mathcal{F}_{j-1}] \alpha_j}{R_{j-1} + 1} \right] \leq \mathbb{E} \left[\sum_{j \in H_0(t)} \frac{\alpha_j}{R_{j-1} + 1} \right] \\ &= \mathbb{E}[\text{FDP}^*(t)] \leq \alpha, \end{aligned}$$

where we used α_j and R_{j-1} are $\mathcal{F}_{j-1}$ -measurable, so that $\mathbb{E}[E_j \mid \mathcal{F}_{j-1}] \leq 1$ pulls out. $\square$

Since $H_0(t) \subseteq \{1, \dots, t\}$ and all terms $\alpha_j/(R_{j-1} + 1) \geq 0$,

$$FDP^*(t) \leq \widehat{FDP}_{\text{e-LORD}}(t)$$

almost surely, giving the FDR guarantee.

Risk-aversion allocation rule. The testing levels α_t are set via the *Risk Aversion Investing* (RAI) rule of Zhang et al. (2025):

$$\alpha_t = \omega_t \cdot rw_t \cdot (R_{t-1} + 1), \quad rw_t = \alpha - \sum_{j=1}^{t-1} \frac{\alpha_j}{R_{j-1} + 1}, \quad (22)$$

where rw_t is the *remaining wealth* and $(\omega_t)_{t \in \mathbb{N}} \in (0, 1)$ governs the fraction of remaining wealth allocated to the current test. Initial wealth $rw_1 = \alpha$, $\alpha_1 = \alpha\omega_1$. The sequence ω_t updates as

$$\omega_{t+1} = \omega_t + \omega_1 \phi^{t-R_t} (1 - \delta_t) - \omega_1 \psi^{R_t} \delta_t, \quad (23)$$

with user-chosen parameters $\phi, \psi \in [0, 0.5]$ ensuring $\omega_t \in (0, 1)$.

Algorithm 1 e-LORD

Require: Initial wealth $rw_1 = \alpha$, sequence (ω_t) , parameters ϕ, ψ

```

1:  $R_0 \leftarrow 0$ 
2: for  $t = 1, 2, \dots$  do
3:    $\alpha_t \leftarrow \omega_t \cdot rw_t \cdot (R_{t-1} + 1)$ 
4:   Observe e-value  $e_t$ 
5:   if  $e_t \geq 1/\alpha_t$  then
6:     Reject  $H_t$ ;  $\delta_t \leftarrow 1$ 
7:   else
8:     Accept  $H_t$ ;  $\delta_t \leftarrow 0$ 
9:   end if
10:   $R_t \leftarrow R_{t-1} + \delta_t$ 
11:   $rw_{t+1} \leftarrow rw_t - \alpha_t/(R_{t-1} + 1)$ 
12:  Update  $\omega_{t+1}$  via (23)
13: end for
```

Remark 6.10 (e-LOND as a special case). Setting $\gamma_t = \omega_t \prod_{j=1}^{t-1} (1 - \omega_j)$ in e-LOND recovers $\alpha_t^{\text{e-LOND}} = \alpha_t$ from (22). e-LOND is therefore a particular instantiation of the RAI framework with a static γ_t schedule.

Alpha-death. With $\omega_1 = 0.1$ over $n = 500$, e-LORD and e-SAFFRON exhaust their wealth by $t \approx 50$ (power ≈ 0.002). The canonical value is $\omega_1 = O(1/T)$ (Zhang et al. 2025): for $n = 500$, take $\omega_1 \leq 0.01$.

6.4 e-SAFFRON (e-GAI framework, Zhang et al. 2025)

e-SAFFRON is the *adaptive* variant of e-LORD. It introduces a predictable sequence $(\lambda_t)_{t \geq 1} \subset (0, 1)$ estimating the proportion of true nulls and *discards* candidate nulls whose e-values are too small, reallocating α -wealth toward more informative hypotheses.

The corresponding FDP estimator is

$$\widehat{FDP}_{e\text{-SAFFRON}}(t) = \sum_{j=1}^t \frac{\alpha_j}{R_{j-1} + 1} \cdot \frac{\mathbb{1}_{\{e_j < 1/\lambda_j\}}}{1 - \lambda_j}.$$

The intuition: when $e_j \geq 1/\lambda_j$, the hypothesis is a candidate alternative and is not charged to the FDP estimate; otherwise it contributes with inverse weight $(1 - \lambda_j)^{-1}$.

Theorem 6.11. *Let $(E_t)_{t \geq 1}$ be an e-process and $(\lambda_t)_{t \geq 1}$ a predictable sequence in $(0, 1)$. Then for every $t \geq 1$:*

$$\mathbb{E}[\widehat{FDP}_{e\text{-SAFFRON}}(t)] \geq \mathbb{E}[FDP^*(t)].$$

Proof.

$$\begin{aligned} \mathbb{E}[\widehat{FDP}_{e\text{-SAFFRON}}(t)] &= \sum_{j=1}^t \mathbb{E} \left[\frac{\alpha_j}{R_{j-1} + 1} \cdot \frac{\mathbb{1}_{\{e_j < 1/\lambda_j\}}}{1 - \lambda_j} \right] \\ &\geq \sum_{j \in H_0(t)} \mathbb{E} \left[\frac{\alpha_j}{R_{j-1} + 1} \cdot \frac{\mathbb{E}[\mathbb{1}_{\{e_j < 1/\lambda_j\}} \mid \mathcal{F}_{j-1}]}{1 - \lambda_j} \right]. \end{aligned}$$

Since E_j is an e-variable and λ_j is $\mathcal{F}_{j-1}$ -measurable, Markov gives $\mathbb{P}(E_j \geq 1/\lambda_j \mid \mathcal{F}_{j-1}) \leq \lambda_j$ under $H_{0,j}$, hence $\mathbb{P}(E_j < 1/\lambda_j \mid \mathcal{F}_{j-1}) \geq 1 - \lambda_j$. Therefore

$$\mathbb{E}[\widehat{FDP}_{e\text{-SAFFRON}}(t)] \geq \sum_{j \in H_0(t)} \mathbb{E} \left[\frac{\alpha_j}{R_{j-1} + 1} \right] = \mathbb{E}[FDP^*(t)].$$

□

Combining with Proposition 6.9, e-SAFFRON controls $\text{FDR} \leq \alpha$ at every t .

Remaining-wealth rule. The testing levels use a modified wealth

$$rw_t^{e\text{-SAFFRON}} = \alpha(1 - \lambda) - \sum_{j=1}^{t-1} \frac{\alpha_j \mathbb{1}_{\{e_j < 1/\lambda_j\}}}{R_{j-1} + 1},$$

and $\alpha_t = \omega_t \cdot rw_t^{e\text{-SAFFRON}} \cdot (R_{t-1} + 1)$, with ω_t updated by (23). The rejection rule remains $\delta_t = \mathbb{1}_{\{e_t \geq 1/\alpha_t\}}$.

Setting $\lambda = 0$ recovers e-LORD: e-SAFFRON is therefore the adaptive counterpart of e-LORD.

6.5 Stopped e-BH (Wang, Dandapanthula & Ramdas 2025)

The offline e-BH procedure (Section 6.1) requires all e-values to be available at once. In sequential settings, data arrive over time and one may want to stop the experiment at a data-dependent time while still controlling FDR. The stopped e-BH procedure addresses this.

The key subtlety is the notion of filtration. Suppose G null hypotheses $H_0^1, \dots, H_0^G$ are tested simultaneously, each with its own *local* filtration $(\mathcal{F}_n^g)_{n \geq 0}$ generated from the data specific to hypothesis g . The *global* filtration is $\mathcal{F}_n = \sigma(\mathcal{F}_n^g : g \in [G])$. Let $\mathcal{T}$ denote the set of stopping times w.r.t. $(\mathcal{F}_n)$.

Algorithm 2 e-SAFFRON

Require: Initial wealth $rw_1 = \alpha(1 - \lambda)$, predictable sequence (λ_t) , parameters ϕ, ψ

```
1:  $R_0 \leftarrow 0$ 
2: for  $t = 1, 2, \dots$  do
3:    $\alpha_t \leftarrow \omega_t \cdot rw_t^{\text{e-SAFFRON}} \cdot (R_{t-1} + 1)$ 
4:   Observe  $e_t$ 
5:   if  $e_t \geq 1/\alpha_t$  then
6:     Reject  $H_t$ ;  $\delta_t \leftarrow 1$ 
7:   else
8:     Accept  $H_t$ ;  $\delta_t \leftarrow 0$ 
9:   end if
10:   $R_t \leftarrow R_{t-1} + \delta_t$ 
11:   $rw_{t+1}^{\text{e-SAFFRON}} \leftarrow rw_t^{\text{e-SAFFRON}} - \alpha_t \mathbb{1}_{\{e_t < 1/\lambda_t\}} / (R_{t-1} + 1)$ 
12:  Update  $\omega_{t+1}$  via (23)
13: end for
```

Definition 6.12 (Global e-process). A non-negative adapted process $(M_n^g)_{n \geq 1}$ is a *global e-process* for $\mathcal{P}^g$ on $(\mathcal{F}_n)$ if at any stopping time $\tau \in \mathcal{T}$, M_τ^g is an e-value for $\mathcal{P}^g$: $\sup_{P \in \mathcal{P}^g} \mathbb{E}^P[M_\tau^g] \leq 1$.

A process that is an e-process only w.r.t. its local filtration $(\mathcal{F}_n^g)$ is a *local e-process*. The distinction is crucial: stopping a local e-process at a global stopping time $\tau \in \mathcal{T}$ does *not* in general yield a valid e-value, which would invalidate the FDR guarantee.

Definition 6.13 (Stopped FDR control). An adapted sequence of random sets (R_n) with $R_n \subseteq [G]$ satisfies *level- α stopped FDR control* if

$$\sup_{P \in \mathcal{M}, \tau \in \mathcal{T}} FDR_\tau \leq \alpha,$$

where $FDR_\tau = \mathbb{E}^P[FDP_\tau]$ and $FDP_\tau = FD_\tau / (R_\tau \vee 1)$.

Definition 6.14 (Stopped e-BH). Run e-BH at each time step on the current values of the e-processes: $R_n = eBH_\alpha(M_n^g : g \in [G])$.

Theorem 6.15 (Stopped e-BH, Wang et al. 2025). *Let $(M_n^g)_{n \geq 1}$, $g \in [G]$, be global e-processes for $(\mathcal{P}^g : g \in [G])$. Then $(R_n) = \{eBH_\alpha(M_n^g : g \in [G])\}$ satisfies level- α stopped FDR control.*

Proof. Fix $\tau \in \mathcal{T}$ and $P \in \mathcal{M}$. Each (M_n^g) being a global e-process, the stopped values $(M_\tau^g : g \in [G])$ are valid e-values. Theorem 6.3 (FDR control of offline e-BH under arbitrary dependence) applied to these stopped e-values gives $FDR_\tau \leq \alpha$ for the set $eBH_\alpha(M_\tau^g : g \in [G])$. Since this holds for any τ and P , stopped FDR control follows. $\square$

This reduces the problem to verifying that locally constructed e-processes are in fact global. Wang et al. (2025) identify a Markovian causal condition under which this holds.

Definition 6.16 (Markovian assumption, Wang et al. 2025, Assumption 3.1). Consider observations $Z_n = (X_n, Y_n)$ where $X_n \in \mathcal{X}$ is a covariate and $Y_n = (Y_n^1, \dots, Y_n^G) \in \mathcal{Y}^1 \times \dots \times \mathcal{Y}^G$ a response vector. The Markovian assumption is

$$Y_n \perp (X_1, \dots, X_{n-1}; Y_1, \dots, Y_{n-1}) \mid X_n \quad \forall n \geq 2.$$

Given the current covariate X_n , Y_n is independent of the past, with arbitrary cross-component dependence in Y_n allowed.

Under this assumption, any e-process built multiplicatively from stepwise e-values computed within a single hypothesis is globally valid (Wang et al. 2025, Theorem 3.3), covering likelihood-ratio and universal-inference e-processes.

Application to jump detection. In our setting, the Markovian assumption is satisfied by the causal window of Section 4.3: $\hat{\sigma}_i$ is $\mathcal{F}_{i-1}$ -measurable (does not look at r_i), so E_i is a global e-value. Complexity is $O(t \log t)$ per step.

6.6 Baselines BH-LM and BH-BNS

BH-LM. `bh_lee_mykland` (`baselines.py` lines 39–58) computes $L_i = r_i/(\hat{\sigma}_i\sqrt{\Delta})$, the two-sided p-value $P_i = 2\mathbb{P}(|Z| > |L_i|)$, then applies BH. Decisions are made per hypothesis.

BH-BNS. In `detector.py` (lines 170–177), BH-BNS calls `bh_bns_global` (global BNS ratio test, `baselines.py` lines 104–120) and duplicates its result across all hypotheses: if the global test rejects, all returns are flagged as jumps; otherwise none. This matches the BS 2016 usage (BNS as a day-level filter).

6.7 Synthetic comparison

Table 2: Summary: guarantees, assumptions, complexity.

Procedure	Guarantee	Dependence	Key parameter	Complexity
e-BH	$\text{FDR} \leq \alpha$	Arbitrary	—	$O(n \log n)$
e-LOND	$\text{FDR} \leq \alpha$	Arbitrary	C_{JM}	$O(1)/\text{step}$
e-LORD	$\text{FDR} \leq \alpha$	Arbitrary	ω_1, ϕ, ψ	$O(1)/\text{step}$
e-SAFFRON	$\text{FDR} \leq \alpha$	Arbitrary	ω_1, λ	$O(1)/\text{step}$
Stopped e-BH	Anytime $\text{FDR} \leq \alpha$	Markov / causal	—	$O(t \log t)/\text{step}$
BH-LM	$\text{FDR} \leq \alpha$	PRDS	K_{win}	$O(n \log n)$
BH-BNS	$\text{FDR} \leq \alpha$	PRDS	—	$O(n)$

7 Software architecture and pipeline

7.1 Five-layer structure

simulate → estimators → values → fdr → pipeline

7.2 Metrics

$$\text{FDP} = \frac{|\mathcal{R} \cap \mathcal{H}_0|}{\max(|\mathcal{R}|, 1)}, \quad \text{Power} = \frac{|\mathcal{R} \cap \mathcal{H}_1|}{\max(|\mathcal{H}_1|, 1)},$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \quad \text{Delay} = \frac{1}{|\mathcal{R} \cap \mathcal{H}_1|} \sum_{i \in \mathcal{R} \cap \mathcal{H}_1} (t_i - t_{i, \text{true}}).$$

Reading the metrics. Each hypothesis $H_{0,i}$ targets a single tick i ; the simulator provides the ground truth $\mathcal{H}_1$ (true jump indices). Three path-level quantities complement FDR and power:

- **Precision** = $|\mathcal{R} \cap \mathcal{H}_1| / \max(|\mathcal{R}|, 1)$: fraction of declared jumps that are genuine — the per-path analogue of $1 - \text{FDP}$.
- **Recall** = $|\mathcal{R} \cap \mathcal{H}_1| / \max(|\mathcal{H}_1|, 1)$: fraction of true jumps that are detected — the per-path analogue of power.
- **F1** = $2 \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$: harmonic mean of the two; the natural scalar summary when precision and recall trade off against each other.
- **Delay**: average number of ticks between the true jump index and the rejection index, over correctly detected jumps. Since each hypothesis $H_{0,i}$ is decided at tick i (one-to-one correspondence), the delay is zero whenever the procedure rejects the exact tick of the jump and positive only if a neighbouring tick is rejected instead. Offline procedures (e-BH, BH-LM) are always zero-delay; online procedures may differ depending on the volatility window.

8 Empirical results

8.1 Protocol

Source: `results/grid_medium.parquet`

Table 3: Parameters of the `grid_medium`.

Parameter	Value
n_{steps}	500
Frequencies (Δt , s)	[5, 60, 300]
Jump regimes	[rare, moderate, hawkes_dense]
Jump sizes (σ)	[3, 5]
α	[0.05, 0.10]
ω_1 (e-LORD/e-SAFFRON)	[0.1, 0.5]
M replications	100
Estimator	MedRV
Algorithms	7 (see Table 1)
Total tasks	32 400
Duration	≈ 7 min

FDR violation convention. For $\alpha = 0.05$, $M = 100$: $\text{SE} = \sqrt{\alpha(1 - \alpha)/M} = \sqrt{0.05 \times 0.95/100} \approx 0.0218$. Violation threshold: $\alpha + 2 \text{SE} \approx 0.094$.

8.2 Result 1: BH's FDR violation at high frequency

Table 4: Empirical FDR per algorithm $\times$ frequency ($\alpha = 0.05$, all regimes, $M = 100$, $n = 500$). Source: `grid_medium.parquet`.

Algorithm	$\Delta t = 5\text{ s}$	$\Delta t = 60\text{ s}$	$\Delta t = 300\text{ s}$	Violates?
BH-BNS	0.096	0.022	0.002	Yes
BH-LM	0.096	0.022	0.002	Yes
Stopped e-BH	0.016	0.005	0.000	No
e-BH	0.010	0.003	0.000	No
e-LOND	0.001	0.000	0.000	No
e-LORD ($\omega_1 = 0.1$)	0.000	0.000	0.000	No (alpha-death)
e-SAFFRON ($\omega_1 = 0.1$)	0.000	0.000	0.000	No (alpha-death)

Violation threshold: $0.094 = \alpha + 2\text{ SE}$

Interpretation. BH's FDR violation at 5s results from BV's bias under dense jumps: BV under-estimates the spot volatility when products $|r_{i-1}||r_i|$ are contaminated, inflating the normalised statistic and producing false positives. This mechanism is structural — it concerns the BV normalisation, not additive noise.

Figure 5 illustrates the evolution of empirical FDR by frequency for $\alpha = 0.05$ and $\alpha = 0.10$.

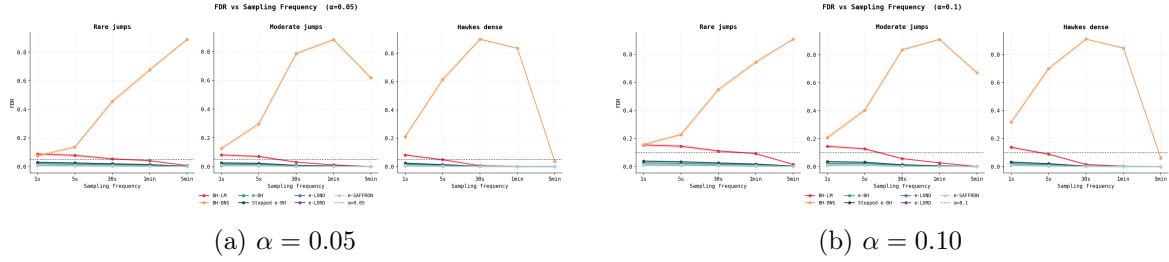


Figure 5: Empirical FDR vs. sampling frequency, by algorithm and regime (`grid_medium`, $M = 100$). The dashed line marks the nominal level α . BH-LM and BH-BNS exceed α at 5s in all regimes.

8.3 Result 2: e-value-based FDR control at all frequencies

E-value-based procedures control the FDR at all tested frequencies (stopped e-BH ≤ 0.016 , e-BH ≤ 0.010 , e-LOND ≤ 0.001) under the three jump regimes, demonstrating robustness to the temporal dependence of Hawkes jumps.

8.4 Result 3: power-safety trade-off

Table 5: Power per algorithm $\times$ frequency ($\alpha = 0.05$, all regimes). Source: `grid_medium.parquet`.

Algorithm	$\Delta t = 5\text{ s}$	$\Delta t = 60\text{ s}$	$\Delta t = 300\text{ s}$
BH-BNS / BH-LM	0.441	0.358	0.157
Stopped e-BH	0.349	0.258	0.028
e-BH	0.326	0.252	0.027
e-LOND	0.235	0.168	0.013
e-LORD ($\omega_1 = 0.1$)	0.002	0.001	0.000
e-SAFFRON ($\omega_1 = 0.1$)	0.002	0.001	0.000

Among procedures that actually control the FDR, **stopped e-BH** is the best compromise (power 0.349 vs. e-BH 0.326, e-LOND 0.235). The loss relative to BH (uncontrolled) is 9–21 pp at 5 s.

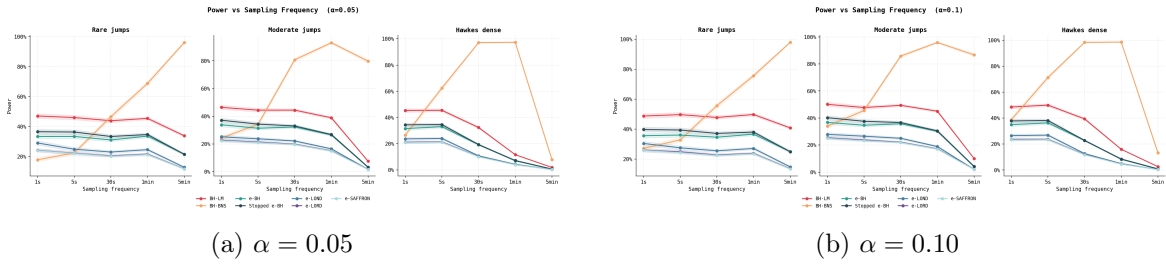


Figure 6: Power vs. sampling frequency by algorithm and regime. Stopped e-BH dominates e-BH and e-LOND at all frequencies among valid procedures. e-LORD/e-SAFFRON with $\omega_1 = 0.1$: alpha-death at 5 s.

8.5 Result 4: FDR-power frontier

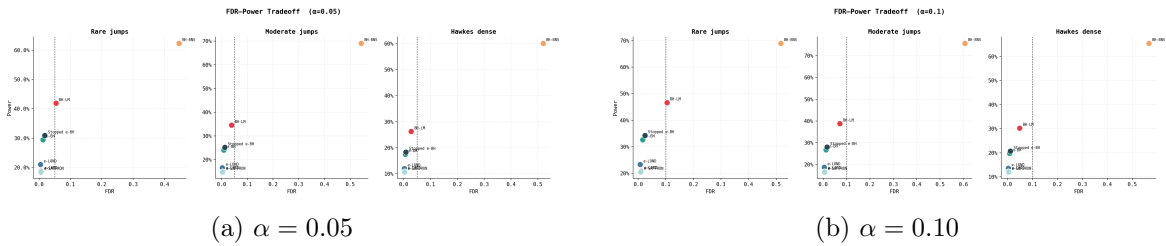


Figure 7: Scatter of empirical FDR vs. power, by algorithm and regime. The dashed vertical line marks the nominal α . BH procedures sit to the right (uncontrolled FDR, high power); e-value procedures sit to the left (controlled FDR, lower power).

8.6 Result 5: Hawkes intensity and jump clustering

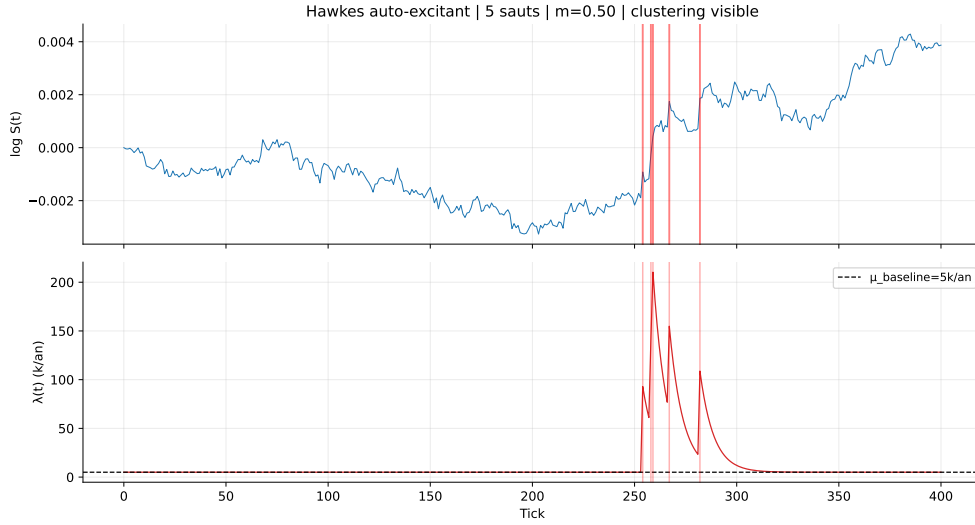


Figure 8: Simulated Hawkes intensity λ_t and corresponding jump arrivals on a single path ($\Delta t = 5$ s, hawkes_dense regime). Bursts of intensity reflect the self-exciting mechanism: each jump raises λ_t , generating temporal clustering.

8.7 Result 6: detailed Hawkes dense regime

Table 6: Detailed results: hawkes_dense, $\Delta t = 5$ s, $\alpha = 0.05$, $M = 100$.

Algorithm	Obs. FDR	Power	F1	FDR controlled?
BH-BNS	0.091	0.423	0.522	Borderline (< 0.094)
BH-LM	0.091	0.423	0.522	Borderline
Stopped e-BH	0.017	0.303	0.454	Yes
e-BH	0.009	0.280	0.435	Yes
e-LOND	0.000	0.204	0.371	Yes
e-LORD ($\omega_1 = 0.1$)	0.000	0.001	0.151	Yes (alpha-death)
e-SAFFRON ($\omega_1 = 0.1$)	0.000	0.001	0.151	Yes (alpha-death)

BH-BNS at 0.091 is below the violation threshold for hawkes_dense alone, but exceeds 0.094 once aggregated across all regimes (0.096). The temporal Hawkes dependence does not aggravate BH’s violation — the violation is inherent to the 5 s frequency, not to clustering.

8.8 Result 7: effect of jump size

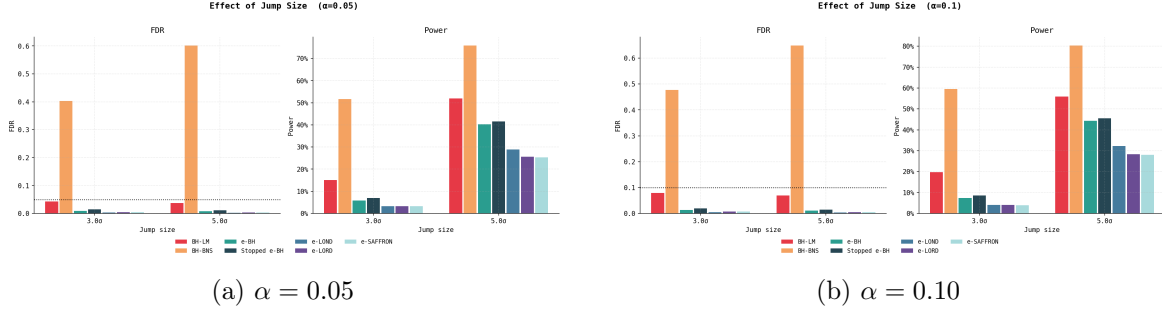


Figure 9: FDR and power as a function of jump size (3σ vs. 5σ). Power increases with jump size for all procedures. BH's FDR remains above α irrespective of jump size at 5s.

8.9 Result 8: heatmap view of aggregated performance

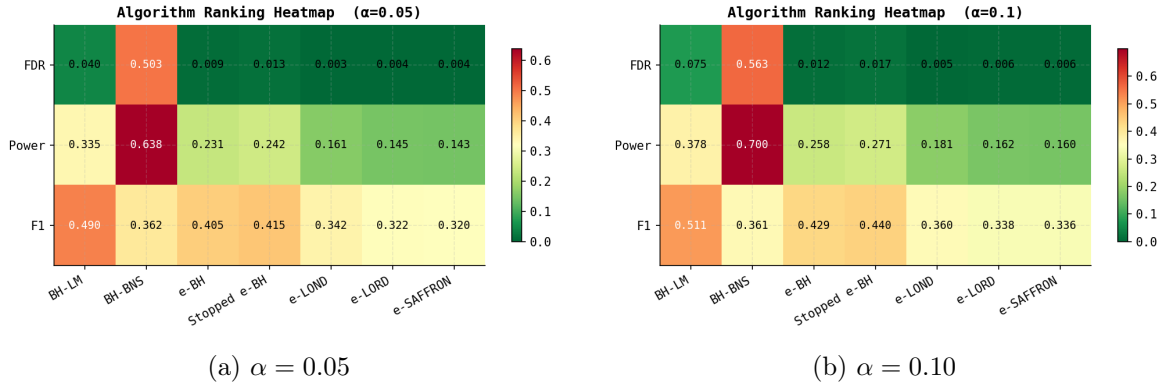


Figure 10: Heatmap of FDR, power, and F1 metrics by algorithm (aggregated across regimes and frequencies). Darker colours mark higher values. BH has the highest power but also the highest FDR. Stopped e-BH offers the best F1 among valid procedures.

8.10 Result 9: BH violation and FDR — walkthrough views

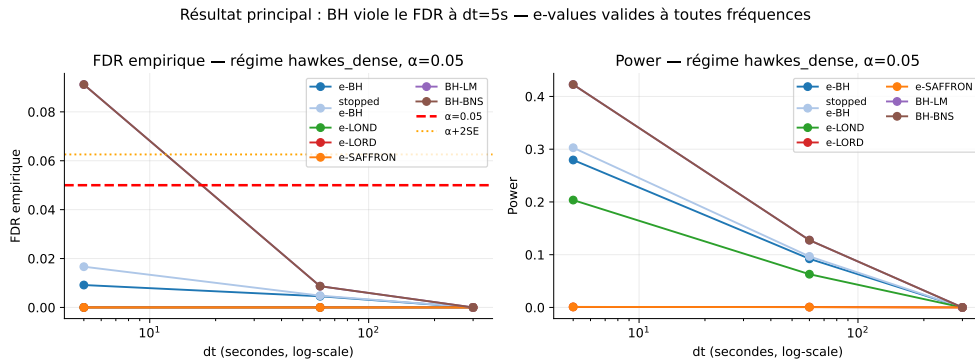


Figure 11: Zoom on the BH FDR violation at $\Delta t = 5$ s: observed FDR per regime and jump size. The dashed line marks $\alpha + 2SE = 0.094$. BH-LM and BH-BNS exceed the threshold in the moderate and hawkes_dense regimes.

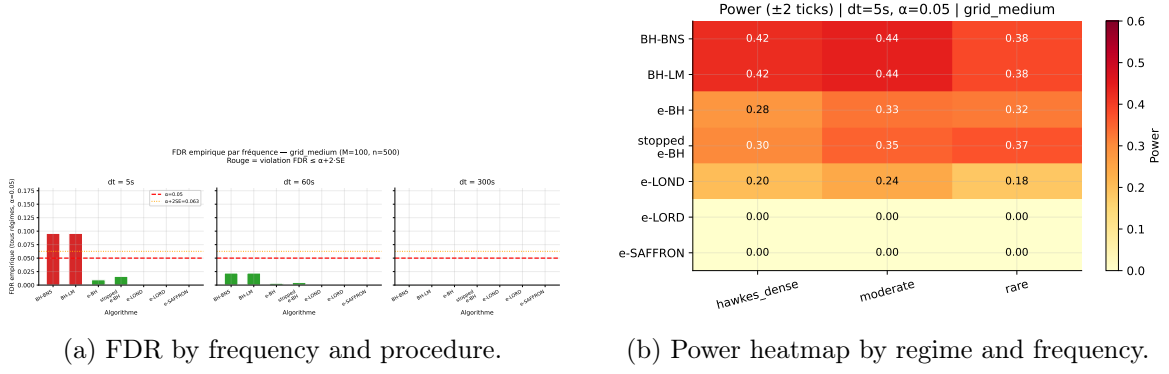


Figure 12: Additional walkthrough views from `00_project_walkthrough.ipynb`, complementing Figures 5 and 10.

8.11 Result 10: alpha-death and parameter ω_1

Alpha-death refers to the exhaustion of wealth $rw_t \rightarrow 0$ for e-LORD and e-SAFFRON. With $\omega_1 = 0.1$ over $n = 500$, both procedures yield power ≈ 0.002 (near zero): rw_t vanishes by $t \approx 50$.

The canonical value recommended by Zhang et al. 2025 is $\omega_1 = O(1/T)$. For $n = 500$, this corresponds to $\omega_1 \leq 0.01$. The results presented with $\omega_1 = 0.1$ illustrate the alpha-death phenomenon; properly calibrated values would restore power.

8.12 Result 11: computational performance

Table 7: Wall-clock per algorithm for $n = 2000$.

Algorithm	ms/run
BH-LM	1
e-BH	170
e-LOND	185
e-SAFFRON	174
e-LORD	187
BH-BNS	257
Stopped e-BH	346

BH-LM is by far the fastest (simple normalisation). Stopped e-BH is the slowest ($O(T^2 \log T)$) but remains under 0.5 s for $n = 2000$.

9 Discussion and limitations

9.1 Debatable methodological choices

Symmetric Gaussian prior. The prior $\pi(J) = \mathcal{N}(0, \tau^2)$ favours no direction. If true jumps are systematically negative, an asymmetric prior would have higher e-power.

Window $K = 50$ not optimised. No sensitivity study on K : too small, the variance of $\hat{\sigma}_i$ is high; too large, it mixes different volatility regimes.

tau_factor = 5.0. A reasonable choice consistent with the grid jump sizes, but not derived from e-power optimisation.

SE convention. The convention $SE = \sqrt{\alpha(1-\alpha)/M}$ (used here) is correct for testing $H_0 : \text{FDR} \leq \alpha$. The empirical convention $SE = \sqrt{\hat{p}(1-\hat{p})/M}$ gives a higher threshold (≈ 0.109) and would not flag a violation for BH at 0.096.

9.2 Validity of the results

What is solid. E-value validity ($\mathbb{E}[E_i] \leq 1$ under H_0 , 86 green tests), monotone e-power, dominance over p-to-e, and FDR control of e-value procedures at all frequencies are robustly established.

What is limited by $M = 100$. BH's FDR violation ($0.096 > 0.094$) is at the detection threshold. The 95% CI for BH-LM's FDR at 5 s is $[0.096 \pm 2 \times 0.022] = [0.052, 0.140]$, which includes the upper bound of the non-violation region. A larger M would confirm or refute the violation with greater statistical power.

10 Conclusion

10.1 Synthesis

This project establishes that e-value-based FDR procedures control the false discovery rate at high frequency where classical methods (BH on Lee-Mykland p-values or BNS) fail. The mechanism is structural: BH requires PRDS, a condition violated under dense jumps at 5 s, whereas e-values only require $\mathbb{E}[E_i | \mathcal{F}_{i-1}] \leq 1$, valid under arbitrary dependence.

10.2 Quantified trade-off

At $\Delta t = 5$ s, $\alpha = 0.05$:

- BH: power 0.441, FDR ≈ 0.096 .
- Stopped e-BH: power 0.349 (−21% relative), FDR ≤ 0.016 .
- e-BH: power 0.326, FDR ≤ 0.010 .
- e-LOND: power 0.235 (−47% relative), FDR ≤ 0.001 .

The cost of guaranteed FDR control is 10–21 pp. Stopped e-BH offers the best power-safety compromise.

10.3 Perspectives

- Properly calibrate $\omega_1 = O(1/n)$ for e-LORD and e-SAFFRON, and re-evaluate their performance.
- Compare estimators BV, MinRV, threshold-BV, and MedRV in the grid.
- Apply to real intraday data (Bitcoin, TAQ, LOBSTER) to validate the pipeline outside of simulation.
- Multi-asset extension: co-jump detection (Jacod-Todorov 2009).

A Derivation of the GROW formula

Starting from (19), with $v = \Delta\hat{\sigma}^2$ and $\pi(J) = \mathcal{N}(0, \tau^2)$:

$$E_i = \int_{-\infty}^{+\infty} \exp\left(\frac{Jr_i}{v} - \frac{J^2}{2v}\right) \cdot \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{J^2}{2\tau^2}\right) dJ.$$

Grouping terms in J :

$$-\frac{J^2}{2v} - \frac{J^2}{2\tau^2} + \frac{Jr_i}{v} = -\frac{(J - \mu^*)^2}{2s^2} + \frac{\mu^{*2}}{2s^2},$$

with $s^2 = v\tau^2/(v + \tau^2)$ and $\mu^* = r_i\tau^2/(v + \tau^2)$.

The Gaussian integral gives $\int \frac{1}{\sqrt{2\pi\tau^2}} e^{-(J-\mu^*)^2/(2s^2)} dJ = s/\tau$, hence:

$$E_i = \frac{s}{\tau} \exp\left(\frac{\mu^{*2}}{2s^2}\right) = \sqrt{\frac{v}{v + \tau^2}} \cdot \exp\left(\frac{r_i^2\tau^2}{2v(v + \tau^2)}\right).$$

This is exactly the expression implemented in `evaluations/construct.py` (lines 40–49).

B Javanmard-Montanari sequence for e-LOND

$$\gamma_t = C_{\text{JM}} \cdot \frac{\log \max(t, 2)}{t \cdot e^{\sqrt{\log \max(t, 2)}}}, \quad C_{\text{JM}} = 0.15708906, \quad \sum_{t=1}^{10^6} \gamma_t \approx 1.0000.$$

The normalisation $\sum_{t=1}^{\infty} \gamma_t = 1$ is the budget condition that ensures FDR control. C_{JM} is computed numerically by truncated summation (`fdr/elond.py` line 12).

C Commented code excerpts

C.1 Heston QE scheme (`simulate/heston.py`)

Listing 1: Case $\psi > 1.5$ (high dispersion) in the QE scheme.

```

1 p = (psi - 1.0) / (psi + 1.0)          # mass at 0
2 beta = 2.0 / (m * (psi + 1.0))        # exponential parameter
3 u = min(U[i], 1.0 - 1e-14)
4 if u <= p:
5     vn = 0.0

```

```

6 else:
7     vn = np.log((1.0 - p) / (1.0 - u)) / beta

```

C.2 Causal window (estimators/spot.py)

Listing 2: Window excluding the tested point i.

```

1 for i in range(n):
2     start = max(0, i - K)
3     sub = log_price[start : i + 1] # returns r[start:i] -- r[i]
        excluded
4     if len(sub) < 4:
5         continue
6     sigma_sq[i] = estimator_fn(sub, dt_seconds)

```

C.3 E-value construction (evaluates/construct.py)

Listing 3: GROW formula.

```

1 dv = dt_years * sigma_hat_i**2
2 tau = tau_factor * sigma_hat_i * np.sqrt(dt_years)
3 tau2 = tau**2
4 numerator = dv / (dv + tau2)
5 exponent = r_i**2 * tau2 / (2 * dv * (dv + tau2))
6 e_value = np.sqrt(numerator) * np.exp(exponent)

```

C.4 e-LOND (fdr/elond.py)

Listing 4: Main loop.

```

1 R = 0
2 for t, e_t in enumerate(e_values, start=1):
3     alpha_t = alpha * _gamma(t) * (R + 1)
4     reject = e_t >= 1.0 / alpha_t if alpha_t > 0 else False
5     if reject:
6         R += 1
7     yield reject

```

C.5 Stopped e-BH (fdr/stopped_ebh.py)

Listing 5: Offline e-BH recomputed at each step.

```

1 buffer: list[float] = []
2 for e_t in e_values:
3     buffer.append(e_t)
4     t = len(buffer)
5     ev = np.asarray(buffer, dtype=float)
6     order = np.argsort(ev)[::-1]
7     thresholds = t / (alpha * np.arange(1, t + 1))

```

```

8     qualifying = np.where(ev[order] >= thresholds)[0]
9     if len(qualifying) == 0:
10         yield False
11     else:
12         k_star = int(qualifying[-1]) + 1
13         yield bool(e_t >= t / (alpha * k_star))

```

D Glossary

Alpha-death Exhaustion of the wealth $rw_t \rightarrow 0$ for e-LORD/e-SAFFRON when ω_1 is too large relative to T .

Anytime-validity FDR controlled at every random stopping time τ .

Branching ratio $m = \alpha_H / (\beta_H \times T)$ for a Hawkes process; $m < 1$ implies stationarity.

E-power $\mathbb{E}_{H_1}[\log E]$: criterion of sequential efficiency.

E-variable $E \geq 0$ with $\mathbb{E}[E \mid H_0] \leq 1$.

FDR False Discovery Rate: $\mathbb{E}[|\mathcal{R} \cap \mathcal{H}_0| / \max(|\mathcal{R}|, 1)]$.

FDP False Discovery Proportion: random realisation of the FDR.

GROW Growth-Rate Optimal in the Worst case: e-value maximising e-power under $\mathbb{E}_{H_0}[E] \leq 1$.

PRDS Positive Regression Dependence on a Subset: sufficient dependence condition for the validity of BH.

RAI Risk Aversion Investing: framework of e-LORD/e-SAFFRON.

Self-consistency Property characterising a broad class of FDR-controlling procedures, generalising e-BH.

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